
Can a Non-Linear Knowledge Graph Make a Small-Model Long-Context Reader Approach the Evidence Ceiling?

A Detective-Novel Case Study

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Abstract

1 Large language models read long documents as a *linear* input history: tokens
2 enter a fixed context window in narrative order, so evidence access depends on po-
3 sition as much as on relevance. We ask whether a *non-linear* retrieval system—a
4 knowledge-graph “mind map” over the text—lets a small-context model approach
5 the ceiling of what the evidence itself permits. We freeze one offline knowledge
6 graph per novel across 30 detective novels and 234 multiple-choice questions,
7 and evaluate three graph-navigation protocols against a recent-window baseline,
8 a whole-book compression baseline, a vector-RAG baseline, and a question-only
9 control, all with a fixed 9B local model under a 16K context. The thirty nov-
10 els span two frozen graph-build pipelines, so pooled numbers are descriptive. The
11 de-biased gold oracle—the model shown every official clue paragraph under a pro-
12 tocol that removes option-position artifacts—reaches 54.7%, while the question-
13 only control reaches 40.2% (paired +14.53 pp, $p = 0.0005$). Graph-guided ev-
14 idence expansion reaches 53.8%, statistically indistinguishable from the oracle
15 ($p = 0.912$), even though it reads only a handful of graph-selected paragraphs per
16 question. We further show that an options-first oracle run is an unreliable ceiling:
17 it is correct for **79.4%** of questions whose answer is option D yet only 21.8–40.0%
18 for A–C, an option-position anchoring artifact that a de-biasing protocol removes.
19 More broadly, we view graph navigation as an instance of *non-linear structured*
20 *memory*—a sparse index a model can hop across, so that reasoning depends on
21 structure rather than position. We expect this primitive to matter more as context
22 windows grow toward a million tokens while documents and long-term memory
23 reach orders of magnitude beyond that.

24 **Keywords:** long-context reasoning; knowledge-graph retrieval; non-linear structured memory;
25 small language models; multiple-choice evaluation.

26 1 Introduction

27 An LLM’s context window is a linear history: the model observes the document token by to-
28 ken and, in a fixed-length window, must decide what to attend to. For a novel of roughly
29 half a million characters (about 10^5 tokens)—far larger than a 16K context—the answer to “why
30 did the killer leave the canned tea” may rest in a paragraph near position 0.28 while the model’s
31 effective attention is pinned near the tail [Liu et al., 2024, Hsieh et al., 2024]. The retrieval problem
32 is therefore not “is the evidence present” but “can the reader navigate to it non-linearly”. Retrieval-

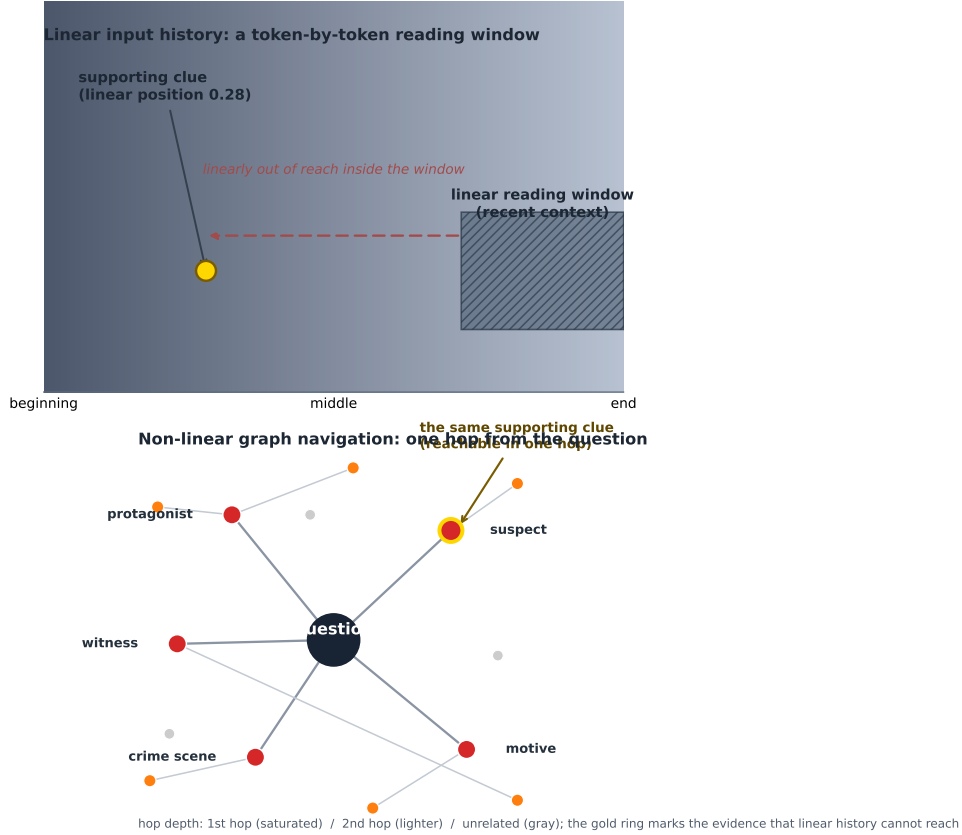


Figure 1: Linear input history vs. non-linear graph navigation. **Top:** the novel is a 0–1 linear reading ruler; a fixed-size reading window on the tail cannot see a supporting clue at position 0.28. **Bottom:** the same clue is one graph hop from the question node, reachable with a tiny context budget independently of linear position. Hop depth is encoded by size and color (1st hop saturated, 2nd hop lighter, unrelated nodes gray) following the dashboard color convention used throughout our analysis.

33 augmented generation [Lewis et al., 2020] mitigates this with a flat similarity index, but it still treats
34 the document as an ordered byte stream to be sliced.

35 This paper studies a different primitive: *graph navigation*. A knowledge graph over the novel—
36 people, locations, clue objects, events, temporal anchors, and the evidence sentences that mention
37 them, connected by typed relations—is a non-linear “mind map”. To answer a question, the model
38 can hop from the query node to a small set of evidence nodes, staying well within a small context
39 budget regardless of where the evidence sits in the linear text (Figure 1). The central question of
40 this paper is: for a small, local, long-context model, can such non-linear graph navigation bring the
41 model close to the *gold ceiling*—the accuracy the model achieves when it is handed every official
42 clue paragraph? The primitive scales beyond this testbed: a 1M-token context is still dwarfed by a
43 100M-token corpus or a long-term memory store, and the bottleneck there is the same linear access
44 we study here, not window size.

45 We construct a clean, frozen benchmark to test this. On 30 detective novels with 234 multiple-
46 choice questions, we fix a 9B local model (16K context, chain of thought disabled) and evalu-
47 ate (i) three graph-navigation protocols built from frozen per-novel graphs, (ii) three standard re-
48 trieval/compression baselines, (iii) a question-only control, and (iv) a *fair gold oracle*: the model
49 shown all official clue paragraphs under a protocol that removes option-position artifacts. We mea-
50 sure whether graph navigation approaches the oracle, and we quantify exactly where the remaining
51 gap comes from.

52 Our headline results are threefold.

- **The de-biased oracle is a real ceiling.** Under a protocol that shuffles options and presents evidence before options, the gold oracle reaches 54.7% versus 40.2% for the question-only control (+14.53 pp, paired exact McNemar $p = 0.0005$); the original options-first gold run reaches only 41.5%, equal to the question-only control (40.2%), because the model anchors on option position.
- **Graph navigation approaches the ceiling.** Graph-guided evidence expansion reaches 53.8%, statistically indistinguishable from the fair gold oracle ($p = 0.912$, paired over the same 234 questions), using only graph-derived evidence. All three graph protocols beat the recent-window baseline on the harder questions.
- **Option-position artifacts must be controlled.** The fair gold oracle’s +14.5 pp advantage is not visible in the options-first oracle run; comparing methods to an un-de-biased oracle is meaningless.

The paper is organized as follows. Section 2 situates the work in long-context and graph-retrieval literature. Section 3 describes the three graph-navigation protocols, the baselines, and the de-biased gold protocol. Section 4 details the frozen corpus and statistics. Section 5 reports main results, the ceiling analysis, the option-position artifact, cohort heterogeneity, and hard-question performance. Section 6 discusses protocol asymmetry and what the results mean for graph retrieval with small models, followed by limitations (Section 7), broader impact (Section 8), and conclusion (Section 9).

2 Related Work

Position bias in long-context models. LLM performance on long inputs degrades with the position of relevant content: models favor content at the beginning and end of long contexts, suffering in the middle [Liu et al., 2024], and similar middle-collapse and recency effects have been reported across model families [Hsieh et al., 2024]. Our recent-window baseline directly measures the recency prior in a clue-dense narrative, and our de-biased oracle removes an orthogonal artifact: option-position anchoring in multiple-choice evaluation.

Graph-based retrieval. GraphRAG [Edge et al., 2024] constructs a community hierarchy over an entity graph and answers queries from aggregated community summaries; HippoRAG [Gutiérrez et al., 2024] builds a personalized PageRank-based retrieval over a knowledge graph, and GraphReader [Li et al., 2024] walks a graph with an agentic controller. These systems optimize for recall on open-ended QA with large frontier models. Our setting differs in three ways: the graphs are frozen offline artifacts (no per-query graph construction), the reader is a fixed 9B local model with a 16K context, and the outcome is a multiple-choice accuracy against a de-biased evidence ceiling rather than open-ended generation.

Hierarchical and structured memory. RAPTOR [Sarathi et al., 2024] builds a tree of recursively summarized clusters for long-context retrieval, and LongRAG [Zhao et al., 2024] retrieves long “group units” to push evidence into the context window. These are linear-to-tree transformations of the text; our graphs additionally encode typed relations and entity resolution, and we compare retrieval protocols against an explicit gold-evidence ceiling.

Long-context QA datasets. DetectiveQA [Xu et al., 2024] releases detective-novel QA with annotated evidence; our benchmark follows its question style and uses its clue-position notion for the gold oracle, but fixes the reader model and freezes the graph artifacts to enable controlled statistical comparison.

3 Method

3.1 Graph representation

For each novel we freeze an offline knowledge graph: nodes are typed entities (*person, location, clue object, event, time anchor*) and *evidence sentences* (paragraphs that mention an entity), and edges are typed relations (*appears in, participates in, temporal before/after, causes, supports, contradicts, related to*). Graphs are built from the novel text without access to future questions or gold answers. Each question is grounded in this graph by matching its entities to nodes, giving a query node from which all graph protocols start.

103 3.2 Graph-navigation protocols

104 All three protocols answer the same multiple-choice question using only graph-derived text, with
105 the original option order, and with no access to baselines or gold.

106 **Graph-guided evidence expansion.** The reader expands from the query node along graph edges
107 to a small set of evidence chunks (order of a handful of supporting paragraphs), concatenates them
108 with the question and options, and answers. This is the “tight” graph expansion protocol: the context
109 is tiny and entirely graph-selected.

110 **Graph-native chunk reranking.** The reader first builds a candidate pool of graph-neighbor chunks
111 (order of a few dozen), then re-ranks the pool using graph-native metadata (hop proximity and entity
112 overlap) to select the top handful of chunks, and answers from those.

113 **Graph-based disagreement arbitration.** The reader runs the first two protocols as independent
114 graph readers; when their selected answers disagree, a graph-only referee decides the final answer
115 from the union of their evidence.

116 3.3 Baselines and control

117 We compare against three standard non-graph conditions and one control, all with the same fixed
118 model and context budget:

- 119 • **Recent-window baseline.** The reader receives the tail of the novel up to the context limit—
120 the natural linear-history baseline.
- 121 • **Whole-book compression baseline.** The novel is compressed into the context window via
122 extractive-compression text, then read.
- 123 • **Vector RAG baseline.** A flat dense retriever selects chunks by question similarity (standard
124 retrieval-augmented generation).
- 125 • **Question-only control.** The reader sees only the question and options—the floor that any
126 evidence-using method must beat.

127 3.4 The de-biased gold oracle and the option-position artifact

128 A gold oracle hands the model every official clue paragraph [Xu et al., 2024] and asks it to answer.
129 We found that this oracle, in its natural *options-first, original-order* form, is useless as a ceiling: the
130 model is **79.4%** accurate when the gold answer is option D but 21.8–40.0% for A–C, and selects D
131 on roughly 59% of all questions. The oracle is learning where the answer sits, not what it means. We
132 therefore define the **fair gold oracle** with a de-biasing protocol: evidence paragraphs are presented
133 *before* the options, and the option order is shuffled per question using a permutation derived deter-
134 ministically from SHA-256 over the question ID, so the model cannot exploit position and every run
135 is reproducible from the question IDs alone. We verify the protocol itself is neutral by re-running
136 the question-only control under the same terse instruction (“Do not explain. Answer with exactly
137 one line: Answer: [letter]”) and shuffled options: it achieves the same 40.2% pooled as the original
138 control (per cohort the two diverge—41.5% vs. 37.2% on the first twenty and 37.1% vs. 47.1% on
139 the final ten—so the pooled match is descriptive), confirming that the +14.5 pp gain of the oracle is
140 evidence, not instruction or order.

141 3.5 Statistics

142 All methods are evaluated on the identical set of 234 questions, so every comparison is a paired
143 comparison. Primary inference uses exact two-sided paired McNemar tests on discordant pairs;
144 we report the delta in accuracy points, wins/losses on discordant pairs, the exact p -value, and a
145 novel-clustered bootstrap 95% confidence interval (5,000 resamples of whole novels, so novels are
146 the sampling unit, not questions). Where a family of graph-vs-baseline contrasts is tested, we ap-
147 ply Holm–Bonferroni correction across the family of nine contrasts (three graph protocols $\times$ three
148 baselines). Deltas are reported as first condition minus second condition (e.g., oracle minus graph
149 method).

Table 1: Main results (micro accuracy, %). “old20”/“new10” are the two frozen cohorts; “Pooled” combines them descriptively. “Hard subset” is accuracy on the 140 questions the question-only control already fails. The de-biased gold oracle is the evidence ceiling.

Method	old20	new10	Pooled	Hard subset	
Fair gold oracle (gold ceiling)	56.7	50.0	54.7%	47.1 (66/140)	
Graph-guided evidence expansion	54.3	52.9	53.8%	42.9	
Graph-native chunk reranking	50.6	51.4	50.9%	45.0	
Graph-based disagreement arbitration	51.2	58.6	53.4%	41.4	Pooled values are
Recent-window baseline	48.2	41.4	46.2%	33.6	
Whole-book compression baseline	51.2	51.4	51.3%	37.1	
Vector RAG baseline	52.4	50.0	51.7%	36.4	
Question-only control	37.2	47.1	40.2%	–	

descriptive: the first 20 and final 10 graphs come from different frozen pipelines. The fair gold oracle is the de-biased upper bound (evidence before options, per-question shuffled options; Section 3.4). The hard subset is the 140 questions the question-only control answers incorrectly (the ones that actually require evidence).

4 Experimental Setup

Corpus. 30 detective novels, 234 multiple-choice questions (164 in the first cohort of 20 novels, 70 in the final cohort of 10). Each question has four options and an official set of clue paragraphs used for the gold oracle.

Model and context. A fixed local 9B model with 16 K context. Chain-of-thought (CoT) prompting [Wei et al., 2022] elicits step-by-step reasoning before an answer is produced; we disable CoT for all conditions so that measured accuracy differences are attributable to the input, not to decoding strategy, and because our task is short multiple-choice answers where CoT mostly changes verbosity rather than content. All 2,808 evaluations (12 conditions $\times$ 234 questions) share this one model: four gold-oracle variants, three graph protocols, three baselines, and two controls.

Frozen graphs and contamination control. Graphs were built offline and frozen; answers never enter the graph. The first 20 and final 10 novels use two different, frozen graph-build pipelines, which we report separately and flag as a descriptive cohort difference: the earlier pipeline used a 7B extractor and a sparser relation schema (mean 456 edges per novel), the later pass used the same 9B model as the reader with a denser schema (mean 758 edges per novel), which is why the two pipelines differ markedly in gold-paragraph recall (Section 5.5). Pooled results across both cohorts are descriptive and labeled as such.

Gold evidence. For each question, the gold evidence is every official clue paragraph with a nonnegative clue position; the final-answer paragraph is excluded. The fair gold oracle concatenates these paragraphs before the (shuffled) options. The option shuffle is seeded per question ID and applied identically in all de-biased runs.

Metrics. Micro accuracy on the 234 questions; per-cohort accuracy; hard-subset accuracy (the 140 questions the question-only control answers incorrectly, i.e. the questions that actually require evidence); per-gold-letter accuracy; and the paired statistics above.

5 Results

5.1 Main results

Table 1 and Figure 2 report pooled accuracy by cohort. All three graph protocols beat the recent-window baseline and the question-only control; graph-guided evidence expansion (53.8%) and graph-based disagreement arbitration (53.4%) are the strongest, matching the fair gold oracle (54.7%).

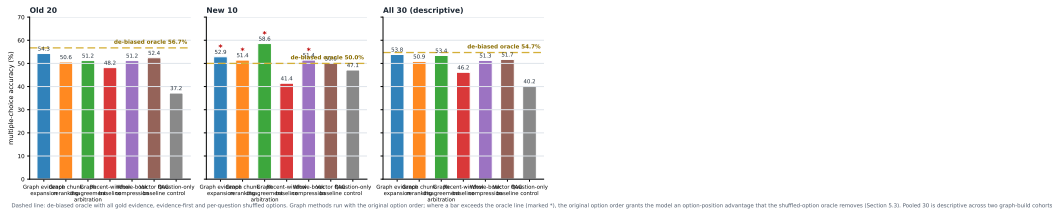


Figure 2: Accuracy by cohort for the seven non-oracle conditions, with the de-biased oracle as a dashed reference line. Bars marked with “*” exceed the oracle line; these run under the original option order while the oracle shuffles options (Section 6), so the position advantage is not exclusive to the graph conditions—the whole-book compression baseline also crosses the line. Pooled values are descriptive across the two graph-build cohorts.

Table 2: External reference points on the old20 cohort (164 questions). “Linear tail” is the final 50,000 characters of the novel. The final row repeats the local 9B graph result for direct comparison.

Condition	Correct/total	Micro acc.
Question-only	68/164	41.5%
Linear tail (50 K chars)	90/164	54.9%
Full novel	132/164	80.5%
Graph-guided evidence expansion (local 9B)	89/164	54.3%

180 5.2 Closed-source reference points

181 To situate the local 9B results, Table 2 reports a closed-source frontier model (DeepSeek v4-flash,
182 chain-of-thought disabled) on the same old20 cohort (164 questions). Given the entire novel text
183 it reaches 80.5%; restricted to a linear 50,000-character tail window it falls to 54.9%; with only
184 the question it reaches 41.5%. Two observations follow. First, linear-history reading caps even a
185 frontier model at 54.9% on this cohort, and only full-novel reading escapes to 80.5%—the cost of
186 linear access. Second, the local 9B model reaches 54.3% through graph navigation alone (Table 1),
187 matching the frontier model’s linear tail-reading accuracy with a small fraction of the context budget;
188 the frontier model’s remaining edge comes from a far larger context window that a 16 K model
189 cannot match. This is a descriptive cross-model reference, not a matched comparison: different
190 models have different intrinsic ceilings, which is precisely why our main analysis compares each
191 model against a model-specific de-biased oracle.

192 5.3 Graph navigation approaches the gold ceiling

193 The central comparison is the gold oracle against the question-only control and the graph proto-
194 cols (Table 3, Figure 3). The fair gold oracle exceeds the question-only control by +14.53 pp (63
195 wins vs. 29 losses on discordant pairs, $p = 0.0005$, 95% CI [6.9, 23.4] pp). This is the evidence
196 effect: the same 9B model, given all official clues under a position-neutral protocol, moves 14.5 ac-
197 curacy points. Graph-guided evidence expansion, given only a handful of graph-derived paragraphs,
198 reaches 53.8%, and the paired contrast against the oracle is +0.85 pp ($p = 0.912$; deltas are reported
199 as first condition minus second, so +0.85 favors the oracle): the graph reader is statistically indistin-
200 guishable from a reader handed the complete gold evidence. Graph-based disagreement arbitration
201 is likewise indistinguishable (+1.28 pp, $p = 0.832$, also favoring the oracle).

202 5.4 Option-position anchoring is a methodological artifact

203 Figure 4 and Table 4 (Appendix) document why an oracle must be de-biased. The options-first
204 gold oracle—all evidence, options first, original order—scores 41.5%, statistically indistinguishable
205 from the question-only control (40.2%), yet this is not a “no evidence helps” result: the oracle is
206 79.4% accurate when the gold answer is D (the gold letter is D on 65 of the 234 questions) but 21.8–
207 40.0% for A–C, at or below the control’s per-letter levels. The 9B model anchors on the position
208 of the options. Shuffling options and moving evidence first removes the artifact: per-letter accuracy

Table 3: Paired contrasts over the same 234 questions (exact two-sided McNemar; W/L = wins/losses on discordant pairs; 95% CI is the novel-clustered bootstrap). The top block shows the gold-oracle analysis; the bottom block shows graph methods against the recent-window baseline with Holm-corrected p .

Paired contrast	Δ	W/L	exact p	Holm p	bootstrap 95% CI
Fair gold vs. question-only (terse)	+14.53pp	63/29	0.0005	–	[6.9, 23.4]
Fair gold vs. graph-guided expansion	+0.85pp	42/40	0.912	–	[-5.9, 8.0]
Fair gold vs. graph arbitration	+1.28pp	46/43	0.832	–	[-6.6, 9.7]
Options-first gold vs. question-only (artifact)	+1.28pp	43/40	0.826	–	[-4.9, 7.8]
Graph-guided expansion vs. recent-window (pooled)	+7.69pp	47/29	0.050	0.454	[2.0, 13.9]
Graph arbitration vs. recent-window (new10)	+17.14pp	15/3	0.008	0.068	[5.4, 31.9]

Exact two-sided paired McNemar over the same 234 questions; W/L = wins/losses on discordant pairs. Bootstrap deltas are novel-clustered (5,000 resamples of whole novels).

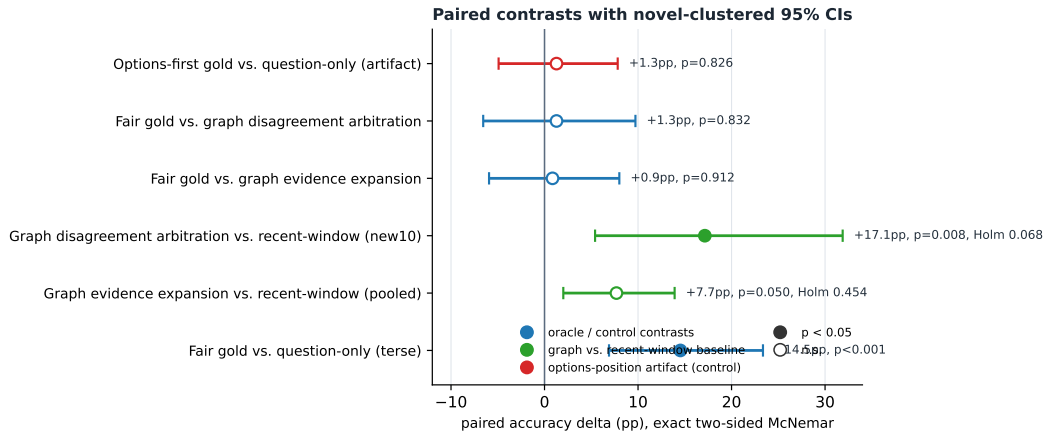


Figure 3: Forest plot of the key paired contrasts. Filled markers are significant at $p < 0.05$; error bars are novel-clustered bootstrap 95% CIs. The headline evidence effect (fair gold vs. question-only) is +14.5 pp, while the graph methods are indistinguishable from the fair gold oracle.

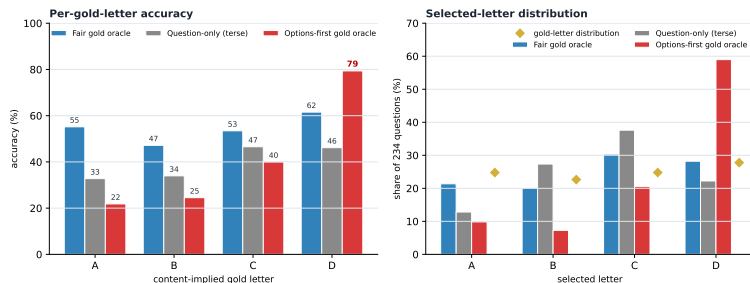
209 flattens (A 55.2%, B 47.2%, C 53.4%, D 61.5% for the fair oracle) and the selected-letter distribution
210 matches the content-implied gold-letter distribution.

211 5.5 Cohort heterogeneity

212 The two frozen cohorts behave very differently, which drives the pooled numbers. In the old20
213 cohort the gold evidence sits overwhelmingly in the graph’s 2-core—the gold-overlap rate is 53.7%
214 versus 18.2% for other nodes (enrichment 2.95, odds ratio 5.21), and graph retrieval reaches only
215 16.3% of gold paragraphs. In the new10 cohort the graph build is far more complete: 73.2% of
216 gold paragraphs are retrieved, the enrichment is 1.61 (odds ratio 2.32), and the best graph protocol
217 reaches 58.6% (versus 50.0% for the de-biased oracle under its more conservative protocol). The
218 two cohorts therefore bound the failure and success modes of the graph pipeline.

219 5.6 Hard-question performance

220 The 140 questions the question-only control gets wrong are the ones that actually require evidence.
221 On this subset the fair gold oracle reaches 47.1% (66/140); graph-native chunk reranking reaches
222 45.0%, graph-guided evidence expansion 42.9%, and graph-based disagreement arbitration 41.4%.
223 Every graph protocol beats the recent-window baseline (33.6%) and the compression (37.1%) and



Options-first presentation lets option position (D) anchor the answer: the options-first gold oracle piles onto D (~59% of questions) and gets D right 79% of the time, while its A/B/C accuracy collapses. De-biasing (evidence first, shuffled options) restores per-letter balance.

Figure 4: **Left:** per-gold-letter accuracy. The options-first oracle is 21.8–40.0% for A–C (at or below the control’s per-letter levels) but 79.4% for D (highlighted), the anchoring signature. **Right:** selected-letter distribution (share of questions); the options-first oracle piles onto D (59%), while the de-biased oracle tracks the content-implied gold-letter distribution.

224 RAG (36.4%) baselines. The gap to the oracle on hard questions is where the graph pipeline loses
 225 evidence, not where the reader fails to reason.

226 5.7 Summary of paired statistics

227 Across the family of nine graph-vs-baseline contrasts, graph-guided evidence expansion vs. recent-
 228 window is the strongest pooled signal (+7.69 pp, $p = 0.050$, Holm-corrected $p = 0.454$); on
 229 the new10 cohort, graph-based disagreement arbitration vs. recent-window is +17.14 pp(15 wins
 230 vs. 3 losses, $p = 0.008$, Holm-corrected $p = 0.068$). The Holm correction is conservative for a
 231 9-comparison family; we report exact and corrected values throughout.

232 6 Discussion

233 **Graph navigation converts position into topology.** The mechanism behind the headline result is
 234 visible in Figure 1: for a small model with a fixed 16 K context, the question is not whether the clue
 235 exists in the book but whether the reader can reach it. A graph hop costs the same budget whether
 236 the target sits at linear position 0.05 or 0.95. Graph-guided evidence expansion uses a few hundred
 237 tokens of graph-selected evidence and matches a reader given the full official clue set—the graph is,
 238 in effect, a non-linear index that lets the model spend its small context where the answer is.

239 **Protocol asymmetry favors the graph methods.** The fair gold oracle shuffles options and presents
 240 evidence first; the graph protocols run the original option order. Shuffling options is harder for a
 241 position-sensitive 9B model, so the oracle’s 54.7% is a conservative ceiling. This is why, on the
 242 new10 cohort, graph methods can exceed the de-biased oracle line (Figure 2): under the original
 243 option order the same evidence is worth more to this model. The asymmetry means our “indistin-
 244 guishable from the ceiling” claim is, if anything, conservative.

245 **Where the graph loses.** The gap between graph methods and the oracle is almost entirely retrieval-
 246 side: on the hard questions the graph protocols reach 41–45% versus 47.1% for the oracle, and
 247 cohort heterogeneity shows the gap is largest precisely where graph recall is lowest (old20: 16.3%
 248 gold recall). Graph-based disagreement arbitration partially recovers lost evidence by running two
 249 readers and arbitrating their disagreement, which is why it is the best protocol on the stronger new10
 250 cohort (58.6%).

251 **Small models are the right testbed.** A frontier model with a large context and strong instruction
 252 following may “search” linearly within its window; a 9B model with 16 K context cannot. Our
 253 results show the graph primitive matters most for exactly this regime—the regime of local, on-device,
 254 cost-sensitive deployment.

255 **Toward non-linear structured memory.** Our graphs instantiate a general primitive: a structured
 256 memory a reader queries by navigation rather than by linear scan. Attention over a long history is
 257 expensive and position-biased; a sparse index trades a small retrieval cost for freedom from position.
 258 The direction scales: a model with a 1M-token context asked to reason over a 100M-token corpus
 259 or long-term memory store faces the same problem a 16 K model faces with a half-million-character

novel—the evidence is somewhere in the history, but the reader cannot afford to see all of it. If navigation is what let the small model approach its evidence ceiling here, the same non-linear index may be what lets much larger models approach their ceilings on histories two orders of magnitude beyond their windows.

7 Limitations

Heterogeneous graph pipelines. The two cohorts use different frozen graph builds; pooled numbers are descriptive, and per-cohort results must be read separately. A single, uniform graph pipeline is needed to make the pooled claim strong.

Exploratory development on the corpus. The graph protocols were developed on these 30 novels; the fixed 9B model and frozen graphs reduce but do not eliminate selection bias. The gold and baseline conditions are independent of the graph development, which protects the ceiling comparison.

Single model. All conclusions are for one 9B checkpoint. The de-biasing protocol and the ceiling comparison should transfer, but the exact magnitudes are model-specific.

Lexical gold matching. Gold evidence is matched to paragraphs by the dataset’s clue positions; the gold oracle sees the official clue set, not necessarily every sentence that could help.

No attention-level causality. We show accuracy differences, not a mechanism; we do not trace attention to graph-selected versus linear evidence.

Contamination nuance. Graphs are built from the full novel, so a graph node may touch the answer region; the graph protocols can surface content beyond the official clue set. This makes the oracle comparison conservative (the oracle is restricted to official clues) but means the graph’s ceiling itself is not bounded by the official clue set.

8 Broader Impact

This is low-risk research on synthetic-style QA over public-domain detective fiction; no user data, no generation of harmful content. The main transferable finding is methodological: *multiple-choice evaluation of small models is sensitive to option order*, and gold/oracle runs that do not shuffle options can produce systematically misleading ceilings (79.4% on D, 21.8–40.0% elsewhere). We recommend that MCQ evaluations of long-context systems shuffle options with a fixed seed and, for evidence ceiling estimation, present evidence before options. The graph-retrieval primitive itself has straightforward applications in document QA where context budgets are tight.

9 Conclusion

We showed that a non-linear knowledge-graph retrieval system lets a small, fixed-context model approach the evidence ceiling on long detective-novel reasoning. Under a de-biased protocol, the gold oracle is a genuine ceiling (+14.5 pp over the question-only control), and graph-guided evidence expansion is statistically indistinguishable from it ($p = 0.912$), while three different graph protocols all outperform linear-history baselines on the questions that require evidence. We also documented an option-position anchoring artifact that makes naive gold oracles unreliable. For small models with tight context budgets, graph navigation is a cheap, effective substitute for linear search. Looking forward, we treat the graph as the simplest form of non-linear structured memory: a sparse index that lets a model spend its attention where the answer is, independent of position. The bottleneck of linear access does not shrink as windows grow, so we expect this primitive to matter more as models move to million-token contexts over corpora and memories of a hundred million tokens or more.

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348 **A Full cohort results**

	Method	correct/total	micro acc. (old20)
	Fair gold oracle (gold ceiling)	93/164	56.7%
	Gold oracle (evidence-first, original order)	97/164	59.1%
	Gold oracle (options-first, shuffled)	88/164	53.7%
	Options-first gold oracle	63/164	38.4%
	Graph-guided evidence expansion	89/164	54.3%
349	Graph-native chunk reranking	83/164	50.6%
	Graph-based disagreement arbitration	84/164	51.2%
	Recent-window baseline	79/164	48.2%
	Whole-book compression baseline	84/164	51.2%
	Vector RAG baseline	86/164	52.4%
	Question-only control (terse)	68/164	41.5%
	Question-only control	61/164	37.2%
	Method	correct/total	micro acc. (new10)
	Fair gold oracle (gold ceiling)	35/70	50.0%
	Gold oracle (evidence-first, original order)	38/70	54.3%
	Gold oracle (options-first, shuffled)	35/70	50.0%
	Options-first gold oracle	34/70	48.6%
	Graph-guided evidence expansion	37/70	52.9%
350	Graph-native chunk reranking	36/70	51.4%
	Graph-based disagreement arbitration	41/70	58.6%
	Recent-window baseline	29/70	41.4%
	Whole-book compression baseline	36/70	51.4%
	Vector RAG baseline	35/70	50.0%
	Question-only control (terse)	26/70	37.1%
	Question-only control	33/70	47.1%
	Method	correct/total	micro acc. (pooled)
	Fair gold oracle (gold ceiling)	128/234	54.7%
	Gold oracle (evidence-first, original order)	135/234	57.7%
	Gold oracle (options-first, shuffled)	123/234	52.6%
	Options-first gold oracle	97/234	41.5%
	Graph-guided evidence expansion	126/234	53.8%
351	Graph-native chunk reranking	119/234	50.9%
	Graph-based disagreement arbitration	125/234	53.4%
	Recent-window baseline	108/234	46.2%
	Whole-book compression baseline	120/234	51.3%
	Vector RAG baseline	121/234	51.7%
	Question-only control (terse)	94/234	40.2%
	Question-only control	94/234	40.2%
	Method	preservation on control-correct (pooled)	
	Graph-guided evidence expansion	70.2%	
	Graph-native chunk reranking	59.6%	
352	Graph-based disagreement arbitration	71.3%	
	Recent-window baseline	64.9%	
	Whole-book compression baseline	72.3%	
	Vector RAG baseline	74.5%	
	Options-first gold oracle	57.4%	

Table 4: Per-gold-letter accuracy (left) and selected-letter distribution (right) for the de-biased and control runs, with the content-implied gold-letter distribution for reference. The options-first oracle’s D spike is an anchoring artifact.

(a) Per-gold-letter accuracy				
Run	A	B	C	D
Fair gold oracle	55.2%	47.2%	53.4%	61.5%
Question-only (terse)	32.8%	34.0%	46.6%	46.2%
Options-first gold oracle	21.8%	24.5%	40.0%	79.4%

(b) Selected-letter distribution				
Run	A	B	C	D
Fair gold oracle	50 (21.4% [†])	47 (20.1% [†])	71 (30.3% [†])	66 (28.2% [†])
Question-only (terse)	30 (12.8% [†])	64 (27.4% [†])	88 (37.6% [†])	52 (22.2% [†])
Options-first gold oracle	23 (9.8% [†])	17 (7.3% [†])	48 (20.5% [†])	138 (59.0% [†])
Gold-letter distribution	58 (24.8% [†])	53 (22.6% [†])	58 (24.8% [†])	65 (27.8% [†])

[†] Column shares are percent of the 234 questions. The options-first gold oracle piles onto D ($138/234 = 59\%$ of questions) regardless of content, while the de-biased fair gold oracle matches the content-implied gold distribution. The options-first oracle refused to select a letter on 8 questions (its row sums to 226).

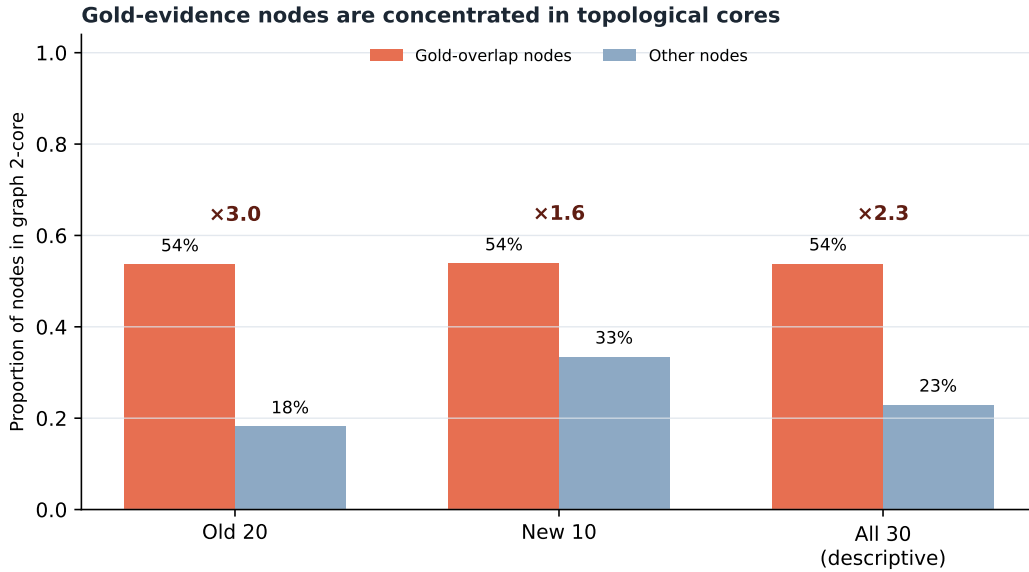
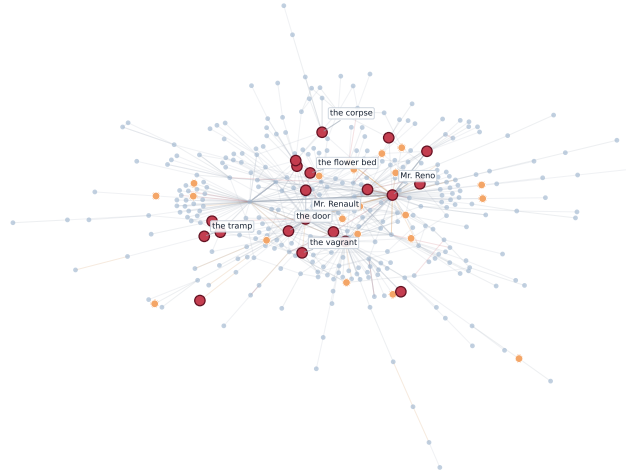


Figure 5: Gold-overlap nodes are concentrated in the topological 2-core of the graph. Rates are micro over all graph nodes per cohort; “x” labels are the gold/other rate ratio (enrichment).

353 B De-biased gold analysis (per-letter and distribution)

354 C Graph structure analysis

Frozen Novel 103 Knowledge Graph



Largest component: 343 nodes, 522 edges; full graph: 396 nodes, 539 edges
Orange: clue-overlap node; dark red with outline: answer-overlap node. Layout distance is topological, not narrative or semantic distance.

Figure 6: Frozen force-layout graph for one novel (n nodes and m edges as annotated). Orange nodes overlap gold evidence; red nodes overlap the answer. The dashed circle marks the topological core. Force-layout distance is topological, not narrative or semantic.

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